# SIMATS ENGINEERING

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SAVEETHA TRANSDISCIPLINARY ANNUAL RESEARCH SUMMIT — STAR SUMMIT 2024 — SIMATS ENGINEERING

# TECH STAR SUMMIT 2024

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# Enhancing Rating Accuracy Through Quantitative Survey on Medication Reminder App for A Comparative Study Between Android Application and iOS Application

## INTRODUCTION

- This Application displays the patient Appointment of the Medication-Reminder application from the doctor login where the doctor can view his daily appointments for clarifications.
- Highlight the significant role that ratings serve in providing objective feedback on the overall application experience, satisfaction, and usability.
- Clearly outlining the objective of the study, which is to enhance the Medication-Reminder's user-centered quality by identifying areas for user experience improvement and measuring user satisfaction using quantitative ratings.
- The study's findings might improve the Medication-Reminder user experience, which would widen the educational and career-growth options available to medical professionals.
- A user-friendly interface, clear appointment notifications, and intuitive features can empower patients to actively engage with their healthcare management, leading to better treatment outcomes and overall health.

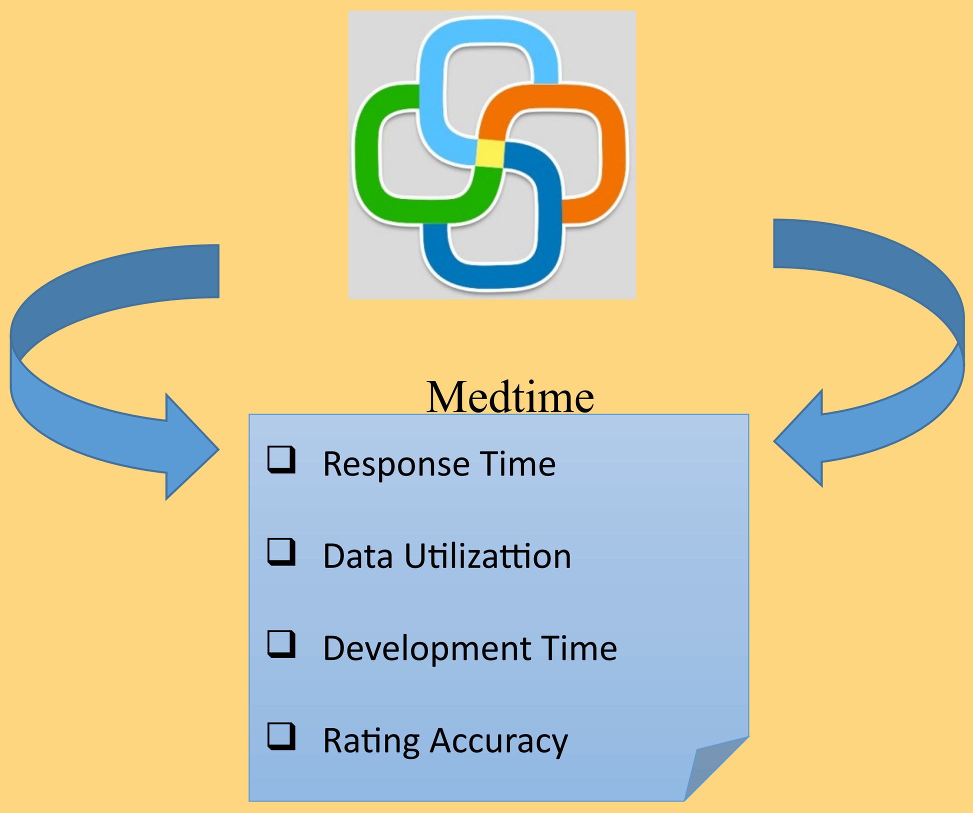


Fig 1:Medication-Reminder

## MATERIALS AND METHODS

Data Collection Processing and measuring the data of sample size of N = 10, N = 10 Total = 20

Choose Application
- Strategy
- Analysis & Planning.
- UI/UX Design.
- App Development.
- Testing.
- Deployment & Support.

Android Application

Android Application Process
- Database – XAMPP(MySQL)
- PHP- Visual Studio code.
- Testing- Postman
- Output- Android Studio

Creating URL for backend connection with database

IOS Application

iOS Application Process
- Database – XAMPP(MySQL)
- PHP- Visual Studio code.
- Testing- Postman
- Output- Swift Xcode

User inputs (GET/POST) for Structure Model & backend connection

Develop and Test the Data

Retrieve User Data from the Database

Retrieve User Data from the Database

Calculating Rating Accuracy

## RESULTS

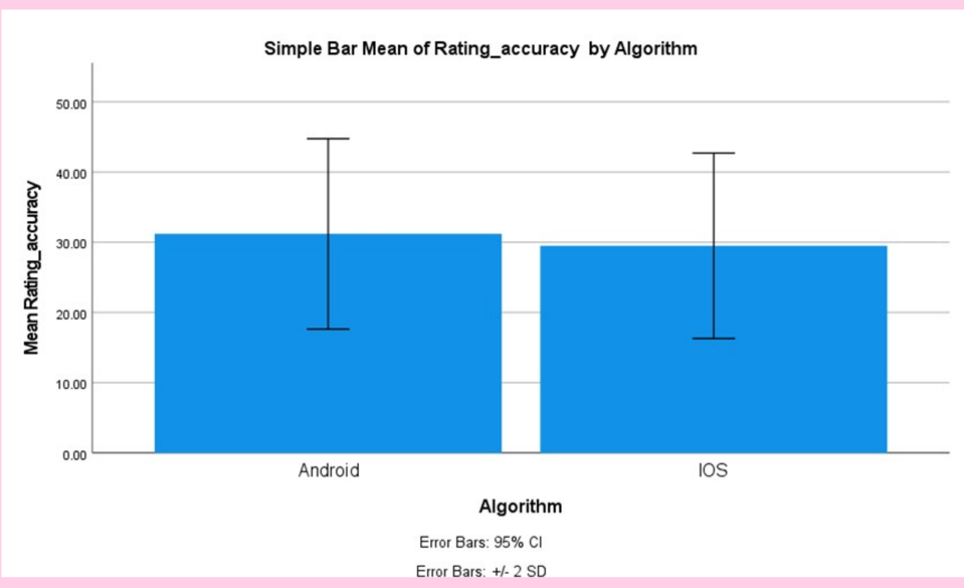


Fig 2: Comparision of Andriod and iOS Application in terms of Rating Accuracy.

**Group Statistics**

| | Algorithm | N | Mean | Std. Deviation | Std. Error Mean |
|---|---|---|---|---|---|
| Rating Accuracy | Android | 10 | 87.67 | 3.310 | .956 |
| | iOS | 10 | 93.00 | 1.954 | .564 |

- Comparision of Andriod and iOS Application in terms of Rating Accuracy.
- The Mean Recovery Rate of iOS is significantly higher than Android in Medication-Reminder.
- This indicates a higher level of precision and reliability in the ratings provided by the iOS application.
- which could be attributed to differences in the underlying algorithms or interface design between the two platforms.

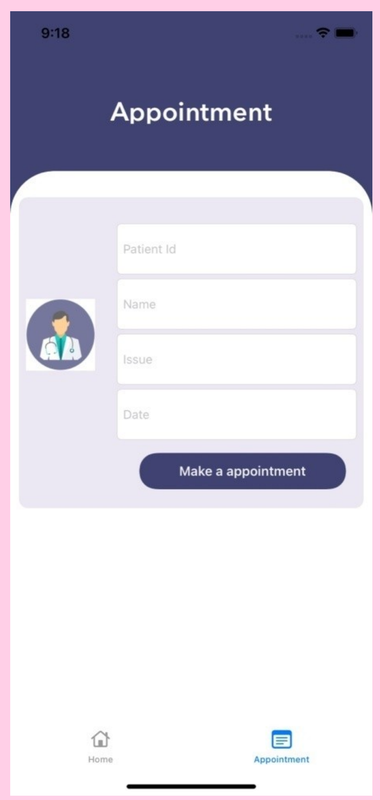


Fig 3:Appointment Booking

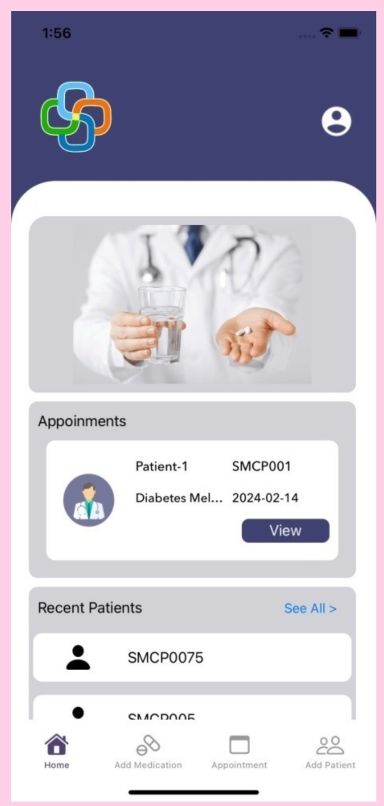


Fig 4:Doctor Home Screen

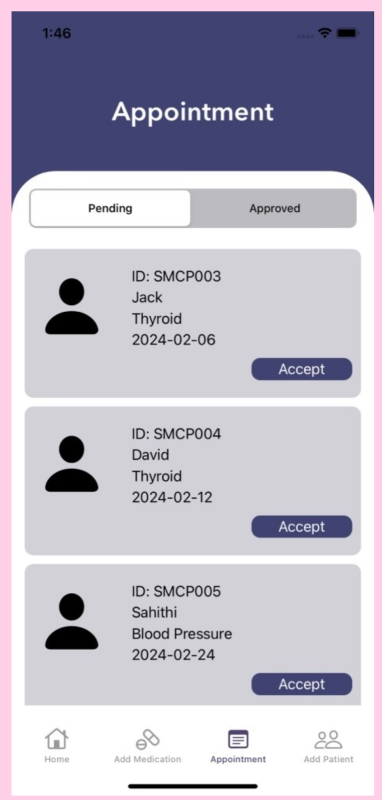


Fig 5:Appointment

## DISCUSSION AND CONCLUSION

- Based on T-test Statistical analysis, the significance value of p=0.01 (independent sample T - test p<0.05) is obtained and shows that there is a statistical significant difference between the group 1 and group 2.
- Overall, the Response time of the Medication-Reminder iOS Application is 89.71ms and it is better than the Android Application.
- Medication-Reminder Android Application is 79.67ms
- The research results clearly show that when it comes to grading accuracy efficiency inside the Medication-Reminder the iOS application performs better than the Android application.
- This suggests that only does the iOS application exhibit superior response time and grading accuracy efficiency, nut it also tends to provide a more satisfactory user experience overall. This aspect could be crucial in determining the overall success and adoption rate of the application among users.

## BIBLIOGRAPHY


- Anderson, Kevin, Oksana Burford, and Lynne Emmerton. 2016. “App Chronic Disease Checklist: Protocol to Evaluate Mobile Apps for Chronic Disease Self-Management.” JMIR Research Protocols 5 (4): e6194.
- Grayek, Emily, Tamar Krishnamurti, Lydia Hu, Olivia Babich, Katherine Warren, and Baruch Fischhoff. 2023. “Collection and Analysis of Adherence Information for Software as a Medical Device Clinical Trials: Systematic Review.” JMIR mHealth and uHealth 11 (1): e46237.
- Salgado, Teresa M., Alexa Fedrigon, Donna Riccio Omichinski, Michelle A. Meade, and Karen B. Farris. 2018. “Identifying Medication Management Smartphone App Features Suitable for Young Adults With Developmental Disabilities: Delphi Consensus Study.” JMIR mHealth and uHealth 6 (5): e9527.
- Tabi, Katarina, Abnashi Singh Randhawa, Fiona Choi, Zamina Mithani, Friederike Albers, Maren Schnieder, Mohammadali Nikoo, et al. 2019. “Mobile Apps for Medication Management: Review and Analysis.” JMIR mHealth and uHealth 7 (9): e13608.